

Introduction

The **bootstrap** approximates the sampling distribution of a statistic by repeatedly resampling from the observed data with replacement and recomputing the statistic each time. Introduced by Efron in 1979, it is a general-purpose tool for standard errors, bias estimates, and confidence intervals when analytical formulas are absent or unreliable.

Prerequisites

Sampling distributions, standard errors.

Theory

Given a sample $\mathbf{X} = (X_1, \dots, X_n)$ and a statistic $\hat{\theta} = T(\mathbf{X})$, the bootstrap:

1. Draws B bootstrap samples $\mathbf{X}^{*(b)}$ of size n with replacement from $\mathbf{X}$.
2. Computes $\hat{\theta}^{*(b)} = T(\mathbf{X}^{*(b)})$ for each.
3. Uses the empirical distribution of $\{\hat{\theta}^{*(b)}\}$ as an estimate of the sampling distribution of $\hat{\theta}$.

From this distribution: SE, bias, percentile or BCa confidence intervals, hypothesis tests.

Validity rests on the bootstrap-principle analogy: the relationship between $\hat{\theta}$ and the true θ is well-approximated by the relationship between $\hat{\theta}^*$ and $\hat{\theta}$.

Assumptions

- iid observations (or an extension for dependent data).
- The statistic is a smooth function of the empirical distribution.

Bootstrap fails for extreme quantiles (e.g., the sample maximum) and for heavy-tailed distributions without finite variance.

R Implementation

```
library(boot)
set.seed(2026)

x <- rgamma(50, shape = 2, rate = 0.5)

# Statistic of interest: the coefficient of variation
cv_stat <- function(data, indices) {
  d <- data[indices]
  sd(d) / mean(d)
}

boot_out <- boot(x, cv_stat, R = 5000)
boot_out

# 95% percentile and BCa CIs
boot.ci(boot_out, type = c("perc", "bca"))

# Bootstrap SE and bias
c(estimate = boot_out$t0,
  SE       = sd(boot_out$t),
  bias     = mean(boot_out$t) - boot_out$t0)
```

Output & Results

ORDINARY NONPARAMETRIC BOOTSTRAP

Call:

```
boot(data = x, statistic = cv_stat, R = 5000)
```

Bootstrap Statistics :

	original	bias	std. error
t1*	0.766	-0.0084	0.0823

Level	Percentile	BCa
95%	(0.612, 0.929)	(0.624, 0.942)

Observed CV = 0.77 with bootstrap SE 0.08. The BCa CI (0.62, 0.94) accounts for skew and bias in the bootstrap distribution.

Interpretation

“The coefficient of variation was 0.77 (bootstrap SE = 0.08; 95 % BCa CI 0.62 to 0.94 based on 5000 bootstrap resamples).”

Practical Tips

- Use BCa intervals by default; they correct for skewness and bias in the bootstrap distribution.
- $B = 1000$ suffices for SEs; $B \geq 10000$ is recommended for accurate BCa CIs.
- For regression, bootstrap cases (rows), residuals, or use the “wild” bootstrap for heteroscedasticity.
- For time-series data, use block bootstrap (`tsboot`).
- The bootstrap can fail: very small samples, heavy tails, or statistics near boundaries (quantiles at 0 or 1).

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests